

NIKHIL REDDY POTTANIGARI

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EXPERIENCES

ServiceNow – Toronto, Canada

APPLIED RESEARCH SCIENTIST

May 2024 – Present

- Led applied research and product engineering for enterprise AI agents, automating document summarization, Q&A, entity extraction, classification, and cross-document reasoning across ServiceNow's Agentic Framework.
- Engineered multimodal retrieval systems uniting OCR text, high-resolution page imagery, and vision-language embeddings to enable multilingual document understanding across complex enterprise formats.
- Partner with NVIDIA research teams on dataset curation, fine-tuning strategies, and standardized benchmarking for document foundation models spanning OCR, retrieval, and document VQA.
- Architected production-grade inference pipelines on Kubernetes using microservices and parallelized sync/async processing for document parsing, visual captioning, and embedding generation.
- Collaborate with cross-functional business units to ship Attachment Summarization and Image Captioning at high enterprise volume, contributing to products associated with approximately \$20M in net-new revenue.
- Extended ServiceNow's Multimodal Services platform to audio and video streams, developing end-to-end ingestion pipelines for multimodal retrieval, QA, and enterprise media analytics.
- Worked on trustworthy AI research, architecting a multi-agent fraud detection system achieving 82% accuracy, and designing red-teaming defenses against prompt injection, jailbreaking, and data exfiltration.

Cyberjustice Laboratory of Udem – Montreal, Canada

GRADUATE STUDENT RESEARCHER

Nov 2023 – May 2024

- Developed JusticeBot AI for legal query processing by fine-tuning LLaMA 2 on Canadian legal corpora via QLoRA; built a specialized RAG pipeline with custom domain embeddings, boosting retrieval accuracy by 40%.
- Co-authored two research papers on LLM evaluation frameworks for legal dispute resolution and published the open-source npm package 'text-moderate' for content safety moderation.

ServiceNow – Hyderabad, India

SOFTWARE ENGINEER - II

June 2021 – Aug 2023

- Architected DocChat, an enterprise semantic search system leveraging BERT embeddings and Pinecone vector database for high-accuracy natural language query processing.
- Engineered an unsupervised recommendation engine using text embeddings, reducing average ticket resolution time by 23%; built the Recommended Actions Framework using React, Java, and GraphQL.

SELECTED PROJECTS

Presence — Self-Evolving Proactive AI Agents

- Co-developed a proactive agent architecture that learns reusable workflows and user preferences from past interactions, deciding when to act, suggest, ask, or stay silent; achieved **~16× context reduction** while retaining strong task performance

Vision Augmented Large Language Models (VA-LLM)

- Built a multimodal AI system integrating CLIP ViT with StableLM via cross-attention for joint vision-language reasoning; containerized MLOps workflow with Hugging Face Hub for reproducible deployment.

System-1 Models for Agentic Decision Making

- Built a benchmark for specialized **System-1 decision models (Jev/Nimble)** against frontier LLMs and confidence-routed hybrids for agent tool selection across **4K+ decisions and 800+ workflows**, evaluating accuracy, calibration, latency, and cost.

PUBLICATIONS

- **Bridging the Modality Gap: Enhancing Document Retrieval with Multimodal Embeddings** — Master's Thesis, MILA / Université de Montréal, 2024
- **Let's Evaluate Step-by-Step: A Robust Evaluation Method of LLM Applications** — *2nd Workshop on Generative AI and Law, GenLaw '24 (Submitted)*
- **Robots in the Middle: Evaluating LLMs in Dispute Resolution** — *JURIX 2024 (Accepted)*
- **Efficient Detection of Disguised Faces from Low-Quality Surveillance Footage** — *IEEE FG 2024 (Accepted)*
- **Text, Pixels, or Both? Evaluating Input Representations for Document QA** — *AIMS workshop COLM 2026 (Accepted)*
- **Preserving What Matters: Semantic Scaffolds Beyond Saturation in Summarization Evaluation** — *CBW workshop EMNLP 2026 (Accepted)*
- **Splitting Documents at Lower Cost: Multi-Split Boundary Decisions for LLM-Based Page Stream Segmentation** — *DocInsights @ EMNLP 2026 (Accepted)*

EDUCATION

MILA - QUEBEC AI INSTITUTE, MONTREAL (Affiliated with UdeM/McGill)

Graduated: Dec 2024

Master's in Computer Science with Machine Learning Specialization; Grade: 4.2/4.3

NATIONAL INSTITUTE OF TECHNOLOGY - WARANGAL

Graduated: May 2021

Bachelor of Technology, Computer Science & Engineering